

ISyE6669 Homework Week 6

Fall 2025

1. Consider the following nonlinear optimization problem:

$$\begin{array}{ll}\min & |2x + 3y| + |2x - 3y| \\ \text{s.t.} & \max\{y, x + y, x - y\} \leq 5.\end{array}$$

Reformulate it as a linear program.

Answer:

The problem can be formulated as a linear program as follows:

$$\begin{array}{ll}\min & z_1 + z_2 \\ \text{s.t.} & -z_1 \leq 2x + 3y \leq z_1 \\ & -z_2 \leq 2x - 3y \leq z_2 \\ & y \leq 5 \\ & x + y \leq 5 \\ & x - y \leq 5 \\ & z_1, z_2 \geq 0\end{array}$$

Solution:

The original problem is a nonlinear optimization problem containing absolute value functions and max functions. To convert this to a linear program, we follow these steps:

1. Introduce auxiliary variables z_1, z_2 to linearize the absolute value functions $|2x + 3y|$ and $|2x - 3y|$.
2. Replace $|2x + 3y|$ with z_1 by adding the constraint $-z_1 \leq 2x + 3y \leq z_1$.
3. Replace $|2x - 3y|$ with z_2 by adding the constraint $-z_2 \leq 2x - 3y \leq z_2$.
4. Decompose the max constraint $\max\{y, x + y, x - y\} \leq 5$ into linear constraints: $y \leq 5$ - $x + y \leq 5$ - $x - y \leq 5$

This transforms the original nonlinear problem into a linear program.

2. Consider the following sets. For each set, determine whether it can be represented using linear inequalities. If it can, provide a formulation using linear inequalities.

- $\{x | \max\{3x + 5, 7x - 3\} \geq 6\}$
- $\{x | \max\{3x + 5, 7x - 3\} \leq 6\}$
- $\{x | |5x - 3| \leq 4\}$
- $\{x | |5x - 3| \geq 4\}$

Answer:

- $\{x | \max\{3x + 5, 7x - 3\} \geq 6\}$: Yes, can be represented as $x \geq \frac{1}{3}$
- $\{x | \max\{3x + 5, 7x - 3\} \leq 6\}$: Yes, can be represented as $x \leq \frac{1}{3}$
- $\{x | |5x - 3| \leq 4\}$: Yes, can be represented as $-\frac{1}{5} \leq x \leq \frac{7}{5}$
- $\{x | |5x - 3| \geq 4\}$: No, cannot be represented using linear inequalities only

Solution:

For each set, we analyze the conditions:

1. $\{x | \max\{3x + 5, 7x - 3\} \geq 6\}$: - We need either $3x + 5 \geq 6$ or $7x - 3 \geq 6$ - $3x + 5 \geq 6 \Rightarrow x \geq \frac{1}{3}$ - $7x - 3 \geq 6 \Rightarrow x \geq \frac{9}{7}$ - Since $\frac{1}{3} < \frac{9}{7}$, the condition reduces to $x \geq \frac{1}{3}$
 2. $\{x | \max\{3x + 5, 7x - 3\} \leq 6\}$: - We need both $3x + 5 \leq 6$ and $7x - 3 \leq 6$ - $3x + 5 \leq 6 \Rightarrow x \leq \frac{1}{3}$ - $7x - 3 \leq 6 \Rightarrow x \leq \frac{9}{7}$ - Since $\frac{1}{3} < \frac{9}{7}$, the condition reduces to $x \leq \frac{1}{3}$
 3. $\{x | |5x - 3| \leq 4\}$: - This is equivalent to $-4 \leq 5x - 3 \leq 4$ - $-4 \leq 5x - 3 \Rightarrow x \geq -\frac{1}{5}$ - $5x - 3 \leq 4 \Rightarrow x \leq \frac{7}{5}$ - Therefore: $-\frac{1}{5} \leq x \leq \frac{7}{5}$
 4. $\{x | |5x - 3| \geq 4\}$: - This is equivalent to $5x - 3 \leq -4$ or $5x - 3 \geq 4$ - $5x - 3 \leq -4 \Rightarrow x \leq -\frac{1}{5}$ - $5x - 3 \geq 4 \Rightarrow x \geq \frac{7}{5}$ - This represents a union of two intervals: $x \leq -\frac{1}{5}$ or $x \geq \frac{7}{5}$ - However, this cannot be represented using linear inequalities only, as it requires a disjunction (OR condition) which is not linear
3. A response variable y is known to depend on a predictor x . A set of n data pairs $\{(x_i, y_i)\}_{i=1}^n$ has been collected. Formulate an optimization model for fitting the “best” straight line

$$y = a + bx$$

to the data, where best is measured with respect to the sum of absolute deviations between observed and predicted values. Can this problem be reformulated as a linear program?

Answer:

Yes, this problem can be reformulated as a linear program. The optimization model is:

$$\begin{aligned} \min \quad & \sum_{i=1}^n z_i \\ \text{s.t.} \quad & -z_i \leq y_i - a - bx_i \leq z_i \quad \forall i = 1, 2, \dots, n \\ & z_i \geq 0 \quad \forall i = 1, 2, \dots, n \end{aligned}$$

Solution:

To minimize the sum of absolute deviations, we introduce variables z_i for each data point that represent the absolute deviation for that point. The objective is to minimize the sum of these absolute deviations.

For each data point (x_i, y_i) , the deviation between the observed value y_i and the predicted value $a + bx_i$ is $|y_i - a - bx_i|$. To minimize the sum of absolute deviations, we minimize $\sum_{i=1}^n |y_i - a - bx_i|$.

By introducing variables z_i and the constraints $-z_i \leq y_i - a - bx_i \leq z_i$ for all i , we ensure that z_i is at least as large as the absolute deviation for point i . Minimizing $\sum_{i=1}^n z_i$ will minimize the sum of absolute deviations.

This formulation is a linear program because: - The objective function is linear in z_i - All constraints are linear in the variables a , b , and z_i - The decision variables are continuous and unrestricted (except for the constraint structure)

4. Hill-O-Beans Coffee Company blends four component beans into three final blends of coffee: One is sold to luxury hotels, another to restaurants and the third to supermarkets for store-label brands. The company has four reliable bean supplies: Argentine Abundo, Peruvian Colmado, Brazilian Maximo and Chilean Saboro. The table below summarizes the very precise recipes for the final coffee blends, the cost and availability information for the four components and the wholesale price per pound of the final blends. The percentages indicate the fraction of each component to be used in each blend.

Component	Hotel	Restaurant	Market	Cost/Pound	Weekly Availability
Abundo	20%	35%	10%	0.60	40,000
Colmado	40%	15%	35%	0.80	25,000
Maximo	15%	20%	40%	0.55	20,000
Saboro	25%	30%	15%	0.70	45,000
Wholesale price/pound	1.25	1.50	1.40		

The processor's plant can handle no more than 100,000 pounds. There is no problem in selling the final blends although the marketing department requires minimum production levels of 10,000, 25,000 and 30,000 pounds respectively for the hotel, restaurant and market blends.

Write a linear program to maximize the weekly profit of Hill-O-Beans. Make sure to explicitly define all decision variables. Implement your model in CVXPY and print out the optimal objective value. What is the optimal amount of each blend which should be produced?

Answer:

Decision Variables:

- X_0 = amount of hotel blend produced (pounds)
- X_1 = amount of restaurant blend produced (pounds)
- X_2 = amount of market blend produced (pounds)
- Y_1 = amount of Abundo component used (pounds)
- Y_2 = amount of Colmado component used (pounds)

- Y_3 = amount of Maximo component used (pounds)
- Y_4 = amount of Saboro component used (pounds)

Linear Programming Formulation:

$$\begin{aligned}
 \max \quad & 1.25X_0 + 1.50X_1 + 1.40X_2 - 0.60Y_1 - 0.80Y_2 - 0.55Y_3 - 0.70Y_4 \\
 \text{s.t.} \quad & Y_1 = 0.20X_0 + 0.35X_1 + 0.10X_2 \\
 & Y_2 = 0.40X_0 + 0.15X_1 + 0.35X_2 \\
 & Y_3 = 0.15X_0 + 0.20X_1 + 0.40X_2 \\
 & Y_4 = 0.25X_0 + 0.30X_1 + 0.15X_2 \\
 & Y_1 \leq 40,000 \\
 & Y_2 \leq 25,000 \\
 & Y_3 \leq 20,000 \\
 & Y_4 \leq 45,000 \\
 & Y_1 + Y_2 + Y_3 + Y_4 \leq 100,000 \\
 & X_0 \geq 10,000 \\
 & X_1 \geq 25,000 \\
 & X_2 \geq 30,000 \\
 & X_0, X_1, X_2, Y_1, Y_2, Y_3, Y_4 \geq 0
 \end{aligned}$$

Solution:

The problem is formulated as a linear program to maximize weekly profit. The objective function represents total revenue from sales minus total cost of components. The constraints ensure:

1. Component-blend relationships based on the recipe percentages
2. Weekly availability limits for each component
3. Plant capacity constraint (total processing limit)
4. Minimum production requirements for each blend
5. Non-negativity constraints

Optimal Solution:

The optimal solution found using Gurobi optimization:

- **Optimal objective value:** \$55,200.00
- **Optimal production amounts:**
 - Hotel blend (X_0): 10,000 pounds
 - Restaurant blend (X_1): 32,500 pounds
 - Market blend (X_2): 30,000 pounds
- **Component usage:**
 - Abundo (Y_1): 16,375 pounds
 - Colmado (Y_2): 19,375 pounds

- Maximo (Y3): 20,000 pounds
- Saboro (Y4): 16,750 pounds

• **Financial summary:**

- Total revenue: \$103,250.00
- Total cost: \$48,050.00
- Total profit: \$55,200.00

5. A farm grows two types of mangoes. This year the yield of mangoes of type 1 is 1000 units and that of type 2 is 700 units. The farm sells all of its mangoes. The mangoes can be sold to a retail company and can also be sold in a local market directly. The contract with the retail company has the following details:

- (a) At least 60% of the total amount of mangoes sold to the retail company must be of type 2.
- (b) The retail company has a demand of 1000 units of mangoes and pays \$3 per unit for both types of mangoes.
- (c) It is possible to sell less than the demand to the retail company: If the total amount of mangoes sold to the retail company is less than 1000 units, then there is a penalty of \$0.05 per unit shortfall from 1000 units. (For example, if the farm decides to sell 900 units of mangoes to the retail company, then the farm earns $900 * 3 - (1000 - 900) * 0.05$)
- (d) It is possible to sell more than the demand to the retail company. However if the number of units of mangoes is more than 1200, then each unit over the first 1200 units is purchased by the retail store at the rate of \$2.5 per unit. (For example, if the farm decides to sell 1300 units of mangoes to the retail company, then the farm earns $1200 * 3 + (1300 - 1200) * 2.5$)

Mangoes of type 1 sell at \$2.2 per unit in the local market and mangoes of type 2 sell for \$3.1 per unit in the local market. Any amount of mangoes can be sold in the local market. Formulate a linear program to maximize the net revenue of the farm.

Answer:

Decision Variables:

- X_{1D} = amount of type 1 mangoes sold directly to local market (units)
- X_{1R} = amount of type 1 mangoes sold to retail company (units)
- X_{2D} = amount of type 2 mangoes sold directly to local market (units)
- X_{2R} = amount of type 2 mangoes sold to retail company (units)

Linear Programming Formulation:

Let $y = X_{1R} + X_{2R}$ be the total amount sold to retail company.

$$\begin{aligned}
\max \quad & z + 2.2X_{1D} + 3.1X_{2D} \\
\text{s.t.} \quad & X_{1D} + X_{1R} = 1000 \\
& X_{2D} + X_{2R} = 700 \\
& X_{2R} \geq 0.6(X_{1R} + X_{2R}) \\
& y = X_{1R} + X_{2R} \\
& z \leq 3.05y - 50 \\
& z \leq 3y \\
& z \leq 2.5y + 600 \\
& y \geq 0 \\
& X_{1D}, X_{1R}, X_{2D}, X_{2R} \geq 0
\end{aligned}$$

Solution:

The problem is formulated as a linear program to maximize net revenue. The revenue from retail company is a piecewise linear function of $y = X_{1R} + X_{2R}$:

1. If $y \leq 1000$: Revenue = $3y - (1000 - y) \times 0.05 = 3.05y - 50$ 2. If $1000 < y \leq 1200$: Revenue = $3y$ 3. If $y > 1200$: Revenue = $3 \times 1200 + 2.5(y - 1200) = 2.5y + 600$

This piecewise linear function is represented using the min function: $\min\{3.05y - 50, 3y, 2.5y + 600\}$.

The variable z represents this minimum value, constrained by: - $z \leq 3.05y - 50$ (first segment) - $z \leq 3y$ (second segment) - $z \leq 2.5y + 600$ (third segment)

The constraints ensure: 1. All type 1 mangoes (1000 units) must be sold either directly or to retail 2. All type 2 mangoes (700 units) must be sold either directly or to retail 3. At least 60% of retail sales must be type 2 mangoes 4. Non-negativity constraints for all variables